



GLOBAL ECONOMICS
GRADE: 11TH
TERM 3 — SOLVED MIDTERM
HYPOTHESIS TEST DESIGN AND DECISION
(C1–C7)
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THIRD
TERM
2025–2026

Learning objective: Design and apply hypothesis tests from real contexts by identifying the parameter, writing hypotheses, selecting the test direction, applying a proportion test and a mean test, interpreting statistical errors, and determining the minimum sample size needed to reject the null hypothesis.

Evaluated criteria:
C1: Describes the concept of a statistical hypothesis.
C2: Identifies the null and alternative hypotheses in a statistical test.
C3: Determines whether a test should be one-tailed or two-tailed.
C4: Applies a hypothesis test in a population proportion setting.
C5: Applies a hypothesis test in a population mean counting setting.
C6: Identifies Type I or Type II error and its consequences.
C7: Finds the minimum sample size needed to reject H_0 .

Critical values used

$$z_{0.95} = 1.645 \quad z_{0.975} = 1.960 \quad z_{0.99} = 2.326$$

For right-tailed tests, reject when $z > z_{\text{critical}}$. For left-tailed tests, reject when $z < -z_{\text{critical}}$. For two-tailed tests, reject when $|z| > z_{\text{critical}}$.

Problem and Tasks

Mandatory requirements

Task outline. Create one consistent real-world counting scenario that compares two groups, such as Prom 2025 vs Prom 2026, Group A vs Group B, before vs after, region 1 vs region 2, two companies, two cohorts, or two markets. The counted variable must be numerical, such as late arrivals, completed assignments, absences, deliveries, or delays.

The same scenario, groups, counted variable, and binary threshold must be used throughout C1–C7.

Required data table format. Show how the data table is organized so that the counting data and the binary transformation can be understood at the same time. Only a few observations are needed to show the structure of the table. After that, the remaining observations can be summarized with the required statistical information.

#	Group A count	Group A binary	Group B count	Group B binary
1	count _{A,1}	0 or 1	count _{B,1}	0 or 1
2	count _{A,2}	0 or 1	count _{B,2}	0 or 1
3	count _{A,3}	0 or 1	count _{B,3}	0 or 1
⋮	⋮	⋮	⋮	⋮

Required summary information. Provide raw data or summarized data for both groups so that the following values are available:

- sample size for each group (n_A, n_B),
- sample mean count for each group ($\bar{x}_A, \bar{x}_B$),
- sample standard deviation for each group (σ_A, σ_B),
- number of successes in each group (x_A, x_B),

- sample proportion for each group ($\hat{p}_A, \hat{p}_B$).

Binary transformation. Define one threshold that converts the counting variable into a binary variable:

$$1 = \text{count above the threshold} \quad 0 = \text{count not above the threshold}$$

C1–C3 setup rule. For C1–C3, students must show **all setup cases**:

1. mean-count setting using Group A as benchmark and Group B as test group,
2. mean-count setting using Group B as benchmark and Group A as test group,
3. proportion setting using Group A as benchmark and Group B as test group,
4. proportion setting using Group B as benchmark and Group A as test group.

C4–C7 calculation rule. For C4–C7, students must choose **one selected benchmark direction** after completing C1–C3. All calculations from C4 to C7 must use only that direction. Students must not calculate C4–C7 for the reverse direction.

For the selected direction, the work must include:

1. the proportion test with the initial sample size, where the test fails to reject H_0 ,
2. the mean-count test with the initial sample size, where the test fails to reject H_0 ,
3. the proportion test with the minimum sample size, where the test rejects H_0 ,
4. the mean-count test with the minimum sample size, where the test rejects H_0 .

Required order. Work in this exact order:

1. **C1.** Present the notation and summary values for both groups in both settings.
 - Mean-count setting: $\mu_A, \mu_B, \bar{x}_A, \bar{x}_B, \sigma_A, \sigma_B, n_A, n_B$.
 - Proportion setting: $p_A, p_B, \hat{p}_A, \hat{p}_B, x_A, x_B, n_A, n_B$.
 - To pass C1, the student must correctly show the data table and identify these values for both groups, such that the initial tests will clearly fail to reject the null hypothesis.
2. **C2.** Write hypotheses for both benchmark directions in both settings.
 - Mean-count setting with Group A as benchmark.
 - Mean-count setting with Group B as benchmark.
 - Proportion setting with Group A as benchmark.
 - Proportion setting with Group B as benchmark.
 - To pass C2, each H_0 and H_a must use the correct parameter, μ for means and p for proportions, such that the initial tests will clearly fail to reject the null hypothesis.
3. **C3.** Identify and justify the tail direction for both benchmark directions in both settings.
 - To pass C3, each setup case must be classified as right-tailed, left-tailed, or two-tailed.
 - Each classification must be justified using the sign in the alternative hypothesis.
 - To pass C3, each setup case must include an approximate hand-drawn normal graph that represents the proposed significance level. The graph does not need to be exact, but the shaded rejection region must be close enough to the stated significance level and must match the correct tail direction.

4. **C4.** Apply the hypothesis test in the **proportion setting** for **one selected benchmark direction**.

- Identify p_0 , $\hat{p}$, and n .
- Calculate the test statistic with the initial sample size and conclude that the test fails to reject H_0 .
- Calculate the test statistic with the minimum sample size found in C7 and conclude that the test rejects H_0 .
- To pass C4, both decisions must be justified with the correct critical value.

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

5. **C5.** Apply the hypothesis test in the **mean-count setting** for **one selected benchmark direction**.

- Identify μ_0 , $\bar{x}$, σ , and n .
- Calculate the test statistic with the initial sample size and conclude that the test fails to reject H_0 .
- Calculate the test statistic with the minimum sample size found in C7 and conclude that the test rejects H_0 .
- To pass C5, both decisions must be justified with the correct critical value.

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

6. **C7.** Find the minimum sample size needed to reject H_0 for **one selected benchmark direction**.

- Find the minimum sample size for the proportion setting.
- Find the minimum sample size for the mean-count setting.
- Use the correct inequality for the selected test direction.
- Round up to the next whole number.
- To pass C7, the minimum sample size must make the test reject H_0 .

$$n = \left(\frac{z\sigma}{\bar{x} - \mu_0} \right)^2 \quad n = \frac{z^2 p_0(1-p_0)}{(\hat{p} - p_0)^2}$$

7. **C6.** Identify the possible error type in context for the four decision cases from **one selected benchmark direction**.

- Proportion setting with initial sample size.
- Mean-count setting with initial sample size.
- Proportion setting with minimum sample size.
- Mean-count setting with minimum sample size.
- To pass C6, each case must state the decision, identify the possible error as Type I or Type II, and explain the error in the context of the scenario.

Reference model: structure your work so it can be solved with the same logic as Problem 1 (Prom 2025 and Prom 2026), using one data source for both mean and proportion settings.

Problem 1 — Grade 11 punctuality audit

The school compares punctuality between Prom 2025 and Prom 2026. The variable is the number of late arrivals to first class per morning.

The binary variable is defined as

1 = more than 9 late arrivals 0 = not more than 9 late arrivals

#	Prom 2025	2025-bin	Prom 2026	2026-bin
1	5	0	5	0
2	6	0	7	0
3	7	0	8	0
4	8	0	9	0
5	9	0	10	1
6	4	0	5	0
7	6	0	8	0
8	7	0	9	0
9	10	1	11	1
10	5	0	6	0
11	8	0	10	1
12	7	0	8	0
13	6	0	7	0
14	9	0	10	1
15	11	1	12	1

#	Prom 2025	2025-bin	Prom 2026	2026-bin
16	5	0	6	0
17	7	0	8	0
18	8	0	10	1
19	10	1	11	1
20	6	0	7	0
21	10	1	10	1
22	7	0	8	0
23	6	0	7	0
24	8	0	10	1
25	11	1	12	1
26	5	0	6	0
27	7	0	8	0
28	9	0	10	1
29	12	1	11	1
30	10	1	9	0

Summary data:

$$n_{25} = 30 \quad \bar{x}_{25} = 7.63 \quad \sigma_{25} = 2.09 \quad x_{25} = 7 \quad \hat{p}_{25} = \frac{7}{30} \approx 0.2333$$

$$n_{26} = 30 \quad \bar{x}_{26} = 8.60 \quad \sigma_{26} = 3.24 \quad x_{26} = 12 \quad \hat{p}_{26} = \frac{12}{30} = 0.40$$

Use $\alpha = 0.05$

C1

Mean setting.

μ_{25} = true mean number of late arrivals per morning for Prom 2025.

μ_{26} = true mean number of late arrivals per morning for Prom 2026.

$$n_{25} = 30 \quad \bar{x}_{25} = 7.63 \quad \sigma_{25} = 3.40$$

$$n_{26} = 30 \quad \bar{x}_{26} = 8.60 \quad \sigma_{26} = 3.24$$

Proportion setting.

p_{25} = true proportion of high-lateness mornings for Prom 2025.

p_{26} = true proportion of high-lateness mornings for Prom 2026.

$$x_{25} = 7 \quad n_{25} = 30 \quad \hat{p}_{25} = \frac{7}{30} \approx 0.2333$$

$$x_{26} = 12 \quad n_{26} = 30 \quad \hat{p}_{26} = \frac{12}{30} = 0.40$$

Using Prom 2025 as benchmark

C2

Proportion test.

$$H_0: p = p_0 \quad H_a: p > p_0$$

$$p = p_{26} \quad p_0 = \hat{p}_{25} = \frac{7}{30} \approx 0.2333$$

$$H_0: p_{26} = 0.2333 \quad H_a: p_{26} > 0.2333 \quad \alpha = 0.05$$

Mean test.

$$H_0: \mu = \mu_0 \quad H_a: \mu > \mu_0$$

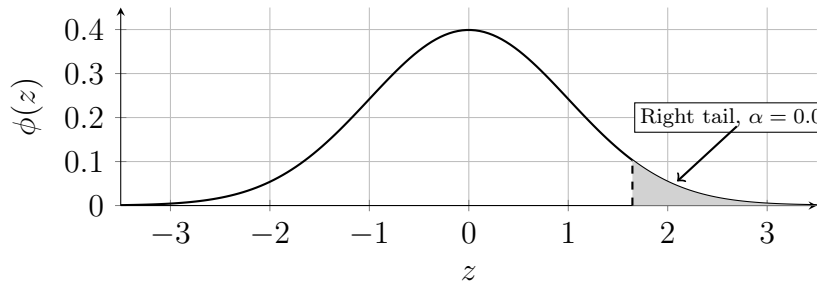
$$\mu = \mu_{26} \quad \mu_0 = \bar{x}_{25} = 7.63$$

$$H_0: \mu_{26} = 7.63 \quad H_a: \mu_{26} > 7.63 \quad \alpha = 0.05$$

C3

Both tests are right-tailed.

$$H_a: p_{26} > 0.2333 \quad H_a: \mu_{26} > 7.63$$



C4

$$\hat{p}_{25} = 0.2333 \quad \hat{p}_{26} = 0.40 \quad n_{26} = 30 \quad k = 2.20$$

$$z = \frac{p_0 - \hat{p}_{25}}{\sqrt{k \frac{\hat{p}_{25}(1-\hat{p}_{25})}{n_{26}}}} = \frac{\hat{p}_{26} - \hat{p}_{25}}{\sqrt{k \frac{\hat{p}_{25}(1-\hat{p}_{25})}{n_{26}}}} = \frac{0.40 - 0.2333}{\sqrt{2.20 \frac{0.2333(0.7667)}{30}}} \\ z = \frac{0.1667}{0.1145} \approx 1.46$$

$$\alpha = 0.05 \quad \longrightarrow \quad z_{\text{critical}} = 1.645$$

$$1.46 < 1.645$$

Fail to reject H_0 . There is not enough evidence to conclude that $p_{26} > 0.2333$.

C5

$$\bar{x}_{25} = 7.63 \quad \bar{x}_{26} = 8.60 \quad \sigma_{25} = 3.40 \quad \sigma_{26} = 3.24 \quad n_{26} = 30$$

The test sample is Prom 2026, so the statistic uses σ_{26} .

$$z = \frac{\bar{x}_{26} - \bar{x}_{25}}{\frac{\sigma_{26}}{\sqrt{n_{26}}}} = \frac{8.60 - 7.63}{\frac{3.24}{\sqrt{30}}}$$

$$z = \frac{0.97}{0.5916} \approx 1.64$$

$$\alpha = 0.05 \quad \longrightarrow \quad z_{\text{critical}} = 1.645$$

$$1.64 < 1.645$$

Fail to reject H_0 . There is not enough evidence to conclude that $\mu_{26} > 7.63$.

C7

Proportion test.

$$\frac{0.1667}{\sqrt{2.20 \frac{0.2333(0.7667)}{n_{26}}}} > 1.645$$

$$n_{26} > \left(\frac{1.645 \sqrt{2.20 \cdot 0.2333(0.7667)}}{0.1667} \right)^2 \approx 38.34$$

$$n_{26, \min} = 39$$

Mean test.

$$\frac{0.97}{\frac{3.24}{\sqrt{n_{26}}}} > 1.645 \Rightarrow n_{26} > \left(\frac{1.645(3.24)}{0.97} \right)^2 \approx 30.18$$

$$n_{26, \min} = 31$$

C6

With the original sample size $n_{26} = 30$, both tests failed to reject H_0 .

Using the minimum sample sizes from C7:

$$z_p = \frac{0.1667}{\sqrt{2.20 \frac{0.2333(0.7667)}{39}}} \approx 1.66 > 1.645$$

$$z_\mu = \frac{0.97}{\frac{3.24}{\sqrt{31}}} \approx 1.67 > 1.645$$

Now both tests reject H_0 .

For the original decision (fail to reject), the possible error is Type II error.

For the recalculated decision (reject), the possible error is Type I error.

Using Prom 2026 as benchmark

C2

Proportion test.

$$H_0: p = p_0 \quad H_a: p < p_0$$

$$p = p_{25} \quad p_0 = \hat{p}_{26} = \frac{12}{30} = 0.40$$

$$H_0: p_{25} = 0.40 \quad H_a: p_{25} < 0.40$$

Mean test.

$$H_0: \mu = \mu_0 \quad H_a: \mu < \mu_0$$

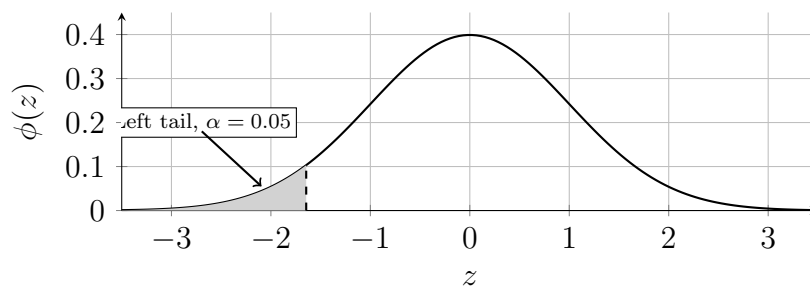
$$\mu = \mu_{25} \quad \mu_0 = \bar{x}_{26} = 8.60$$

$$H_0: \mu_{25} = 8.60 \quad H_a: \mu_{25} < 8.60$$

C3

Both tests are left-tailed.

$$H_a: p_{25} < 0.40 \quad H_a: \mu_{25} < 8.60$$



C4

$$\hat{p}_{26} = 0.40 \quad \hat{p}_{25} = 0.2333 \quad n_{25} = 30$$

$$z = \frac{\hat{p}_{25} - \hat{p}_{26}}{\sqrt{\frac{\hat{p}_{26}(1-\hat{p}_{26})}{n_{25}}}} = \frac{0.2333 - 0.40}{\sqrt{\frac{0.40(0.60)}{30}}}$$

$$z = \frac{-0.1667}{0.0894} \approx -1.86$$

$$z_{\text{critical}} = -1.645$$

$$-1.86 < -1.645$$

Reject H_0 . There is enough evidence to conclude that $p_{25} < 0.40$.

C5

$$\bar{x}_{26} = 8.60 \quad \bar{x}_{25} = 7.63 \quad \sigma_{26} = 3.24 \quad \sigma_{25} = 3.40 \quad n_{25} = 30$$

The test sample is Prom 2025, so the statistic uses σ_{25} .

$$z = \frac{\bar{x}_{25} - \bar{x}_{26}}{\frac{\sigma_{25}}{\sqrt{n_{25}}}} = \frac{7.63 - 8.60}{\frac{3.40}{\sqrt{30}}}$$

$$z = \frac{-0.97}{0.6207} \approx -1.56$$

$$z_{\text{critical}} = -1.645$$

$$-1.56 > -1.645$$

Fail to reject H_0 . There is not enough evidence to conclude that $\mu_{25} < 8.60$.

C7

Proportion test.

$$\frac{0.1667}{\sqrt{1.50 \frac{0.40(0.60)}{n_{25}}}} > 1.645 \Rightarrow n_{25} > \left(\frac{1.645 \sqrt{1.50 \cdot 0.40(0.60)}}{0.1667} \right)^2 \approx 44.54$$

$$n_{25,\min} = 45$$

Mean test.

$$\frac{0.97}{\frac{3.40}{\sqrt{n_{25}}}} > 1.645 \Rightarrow n_{25} > \left(\frac{1.645(3.40)}{0.97} \right)^2 \approx 33.23$$

$$n_{25,\min} = 34$$

C6

With the original sample size $n_{25} = 30$, both tests failed to reject H_0 .

Using the minimum sample sizes from C7:

$$z_p = \frac{-0.1667}{\sqrt{1.50 \frac{0.40(0.60)}{45}}} \approx -1.66 < -1.645$$

$$z_\mu = \frac{-0.97}{\frac{3.40}{\sqrt{34}}} \approx -1.66 < -1.645$$

Now both tests reject H_0 .

For the original decision (fail to reject), the possible error is Type II error.

For the recalculated decision (reject), the possible error is Type I error.